

Lecture #21, Nov 25, 2025

In lecture 20, we discussed Galerkin approximation

Galerkin approximation of PDE

$$- u'(x) = f(x), \quad x \in [0, 1]$$

$$u(0) = \alpha$$

$$u(1) = \beta$$

$$\text{Idea: } u_n(x) = \phi_0(x) + \sum_{j=1}^n a_j \phi_j(x)$$

$$\phi_0(0) = \alpha, \quad \phi_0(1) = \beta, \quad \phi_0(x) \text{ satisfies B.C.}$$

$$\phi_j(0) = 0, \quad \phi_j(1) = 0, \quad \text{for } j = 1, 2, \dots, n$$

$$\{\phi_j(x)\}_{j=1}^n \text{ satisfy zero boundary condition.}$$

$$\text{that means } u_n - \phi_0(x) \in \text{span} \{\phi_j\}_{j=1}^n$$

Today, we begin with strong form vs weak form

Strong form of PDEs

$$- \nabla \cdot (\nu \nabla u) = f \quad \text{in } \Omega$$

$$u = g \quad \text{on } \partial\Omega$$



$$\text{expect } u \in C^2(\Omega) \cap C(\partial\Omega)$$

Call this strong form.

$$\text{Ex: } \Omega = (0, 1) \subset \mathbb{R}, \quad \partial\Omega = \{0, 1\}$$

$$u(x) = x(1-x), \quad u \in C^\infty(0, 1) \Rightarrow u \in C^2(0, 1)$$

$u(x)$ extends to $\{0,1\}$

Weak form

$$\nabla \cdot (\vec{F} \phi) = \vec{F} \cdot \nabla \phi + \phi \nabla \cdot \vec{F}, \quad \begin{array}{l} \phi \text{ is scalar} \\ \vec{F} \text{ is vector} \end{array}$$

Multiply PDE by a "test" function $v(x)$ and integrate

$$-\int_{\Omega} \underbrace{[\nabla \cdot (\vec{F} \phi)]}_{\vec{F}} \underbrace{v}_{\phi} dx = \int_{\Omega} f(x) v(x) dx$$

$$\int_{\Omega} \nabla \cdot (\vec{F} \phi) dx = \int_{\partial \Omega} (\vec{F} \cdot \hat{n}) \phi dx = 0 \quad \text{if } \phi = 0 \text{ on } \partial \Omega$$

div. theorem + B.C

$$= \int_{\Omega} \vec{F} \cdot \nabla \phi dx + \int_{\Omega} \phi \nabla \cdot \vec{F} dx$$

$$0 = \int_{\Omega} [\nabla \cdot (\vec{F} \phi)] v dx = \int_{\Omega} v \nabla u \cdot \nabla v dx + \int_{\Omega} v \nabla \cdot (\vec{F} u) dx$$

Finally

$$\boxed{\int_{\Omega} v \nabla u \cdot \nabla v dx = \int_{\Omega} f v dx}$$

this called the weak form.

Ex: Find weak form of

$$-u''(x) = f(x) \quad 0 < x < 1, \quad \Omega = (0,1)$$

$$u(0) = 0$$

$$u'(1) = 0$$

Solⁿ

let $v(x)$ be any differentiable function s.t $v(0) = 0$

Then

$$-\int_0^1 u''(x) v(x) dx = \int_0^1 f(x) v(x) dx$$

$$-u'(x) v(x) \Big|_0^1 + \int_0^1 u'(x) v'(x) dx = \int_0^1 f(x) v(x) dx$$

$$-u'(1) v(1) + u'(0) v(0) + \int_0^1 u'(x) v'(x) dx = \int_0^1 f(x) v(x) dx$$

define $\langle u, v \rangle = \int_0^1 u v dx$

then $\langle u'(x), v'(x) \rangle = \langle f, v \rangle$

denote $a(u, v) = \langle u'(x), v'(x) \rangle = \int_0^1 u'(x) v'(x) dx$

this is bilinear form

The weak form is

$$\boxed{a(u, v) = \langle f, v \rangle}$$

Theorem

If $u(x)$ satisfies the weak form of a PDE then $u(x)$ is a solution of the strong form.

Big Idea

want to find

$$u_h = \phi_0 + \sum_{j=1}^n c_j \phi_j$$

$$\text{s.t. } a(u_h, \phi_i) = \langle f, \phi_i \rangle \text{ for } 1 \leq i \leq n$$

here we have take $\phi_i(x)$ to be test function $v(x)$

$$\Rightarrow \left. \begin{aligned} a\left(\sum_{j=1}^n c_j \phi_j, \phi_i\right) &= \langle f, \phi_i \rangle \\ \sum_{j=1}^n a(\phi_j, \phi_i) c_j &= \langle f, \phi_i \rangle \end{aligned} \right\} \begin{aligned} a(c_1 \phi_1 + c_2 \phi_2, \phi_1) \\ = c_1 a(\phi_1, \phi_1) + c_2 a(\phi_2, \phi_1) \end{aligned}$$

$$\Rightarrow A c = b, \quad A = [a(\phi_j, \phi_i)], \quad c = [c_j], \\ b = [\langle f, \phi_i \rangle]$$

If A is non singular then this system has a unique solution. And thus u_h will be a solⁿ of the strong form.

How to choose $\{\phi_j\}_{j=1}^n$

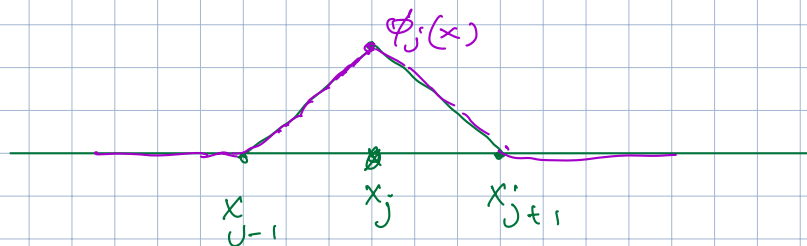
Goal: $\phi_j(x)$ is mostly zero in Ω

1) Simple choice $\phi_1 = x, \phi_2 = x^2, \phi_3 = x^3, \dots$

$$\Omega = (0,1)$$



2) A natural choice is nodal functions.



$$\phi_j(x) = \begin{cases} \frac{x - x_{j-1}^*}{x_j^* - x_{j-1}^*} & x_{j-1}^* \leq x \leq x_j^* \\ \frac{x_{j+1}^* - x}{x_{j+1}^* - x_j^*} & x_j^* \leq x \leq x_{j+1}^* \\ 0 & \text{otherwise} \end{cases}$$

Ex:

$$-u''(x) = f(x) \quad 0 < x < 1$$

$$u(0) = \alpha \quad \leftarrow \text{essential b.c.}$$

$$u'(1) = \beta \quad \leftarrow \text{natural b.c.}$$

$$\text{Take } u_h(x) = c_0 + c_1 x + c_2 x^2 + c_3 x^3$$

$$\text{find the matrix } A = [a(\phi_i, \phi_j)]$$

Sol:

$$\text{We require } \phi_0(0) = \alpha$$

$$\phi_j(0) = 0 \quad \text{for } j = 1, 2, 3, 4$$

$$\text{thus } c_0 = \alpha$$

Take $\beta = 0$
for
Simplicity

$$u_h = \sum_{j=1}^h c_j \phi_j, \quad \phi_1 = x, \quad \phi_2 = x^2, \quad \phi_3 = x^3$$

$$-\int_0^1 u'(x) v(x) dx = \int_0^1 f(x) v(x) dx$$

$$a(u, v) = \langle f, v \rangle, \quad a(u, v) = \int_0^1 u'(x) v'(x) dx$$

$$\begin{aligned} a(\phi_1, \phi_1) &= \int_0^1 \phi_1' \phi_1' dx \\ &= \int_0^1 (x)' (x)' dx \\ &= 1 \end{aligned}$$

$$\begin{aligned} a(\phi_1, \phi_2) &= \int_0^1 \phi_1' \phi_2' dx = \int_0^1 (x)' (x^2)' dx \\ &= \int_0^1 (1) 2x dx \\ &= x^2 \Big|_0^1 = 1 \end{aligned}$$

.....
Complete as H.W.

$$\text{Finally } A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 4/3 & 3/2 \\ 1 & 3/2 & 9/5 \end{bmatrix} \quad Ac = b$$

Remark Let $g = \frac{x^2}{4} \sin \frac{\pi x}{2}$ for the above problem

$$\begin{aligned} b_1 = \langle g, \phi_1 \rangle &= \int_0^1 \frac{x^2}{4} \sin \frac{\pi x}{2} \cdot x dx \\ &= \int_0^1 \frac{x^3}{4} \sin \frac{\pi x}{2} dx \end{aligned}$$

using numerical method find b_1, b_2, b_3

Ex:

$\{x_j\}_{j=0}^n$ is a nonuniform grid, $h_j = x_{j+1} - x_j$
of $[0,1]$

Take nodal basis s.t. $u(0)=0$

Compute the entries of finite element
matrix $[a_{ij}] = [a(\phi_j, \phi_i)]$

When $a(u,v) = \int_0^1 u' v' dx$ is bilinear form.

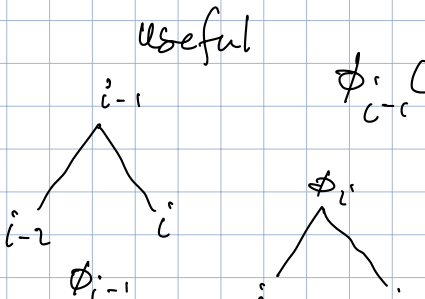
Sol:

$$\phi_i(x) = \begin{cases} \frac{x - x_{i-1}}{x_i - x_{i-1}} & x_{i-1} \leq x \leq x_i \\ \frac{x_{i+1} - x}{x_{i+1} - x_i} & x_i \leq x \leq x_{i+1} \\ 0 & \text{other wise} \end{cases}$$

want $a(\phi_j, \phi_i) = \int_0^1 \phi_j' \phi_i' dx$ $i = 1 \dots n$
 $j = 1 \dots n$

$$a(\phi_{i-1}, \phi_i) = \int_0^1 \phi_{i-1}' \phi_i' dx$$

useful


$$\phi_{i-1}(x) = \begin{cases} \frac{x - x_{i-2}}{x_{i-1} - x_{i-2}} & x_{i-2} \leq x \leq x_{i-1} \\ \frac{x_i - x}{x_i - x_{i-1}} & x_{i-1} \leq x \leq x_i \\ 0 & \text{other wise} \end{cases}$$

$$\begin{aligned} a(\phi_{i-1}, \phi_i) &= \int_0^1 \phi_{i-1}' \phi_i' dx = \int_{x_{i-1}}^{x_i} \phi_{i-1}'(x) \phi_i'(x) dx \\ &= \int_{x_{i-1}}^{x_i} \frac{-1}{x_i - x_{i-1}} \frac{1}{x_i - x_{i-1}} dx \end{aligned}$$

$$= -\frac{1}{h_i^v} \int_{x_{i-1}^*}^{x_i^*} dx$$

$$= -\frac{1}{h_i^v} [x]_{x_{i-1}^*}^{x_i^*}$$

$$= -\frac{1}{h_i^v} \frac{x_i^* - x_{i-1}^*}{1}$$

$$= -\frac{1}{h_i^v}$$

H.W. Compute $\int_0^1 \phi_i'(x) \phi_i'(x) dx$

Lecture #22, Nov 27, 2025

Mass matrix, stiffness matrix, load vector
and time dependent PDE (if time permits)

Consider $-\frac{\partial}{\partial x}\left(\gamma \frac{\partial u}{\partial x}\right) + b u = f(x)$, $u(0) = 0$, $u(1) = 0$

idea: multiply with a test function $v(x)$ and integrate

After using integration by parts

$$\int_0^1 \gamma \frac{\partial u}{\partial x} \frac{\partial v}{\partial x} dx + \int_0^1 b u v dx = \int_0^1 f(x) v(x) dx$$

bilinear linear form

Take $u \approx u_h = \sum_{j=1}^n c_j \phi_j(x)$ and $v = \phi_i$ $i = 1 \dots n$

Upon substitution

$$\gamma \int_0^1 \frac{\partial \phi_j}{\partial x} \frac{\partial \phi_i}{\partial x} dx + b \int_0^1 \phi_j \phi_i dx = \int_0^1 f(x) \phi_i(x) dx$$

Stiffness matrix is $A = [a(\phi_j, \phi_i)]$ $a(\phi_j, \phi_i) = \int_0^1 \phi_j' \phi_i' dx$

Mass matrix $M = [m(\phi_j, \phi_i)]$, $m(\phi_j, \phi_i) = \int_0^1 \phi_j \phi_i dx$

load vector $F = \int_0^1 f \phi_i dx$

FEM system $Ac + Mc = F$

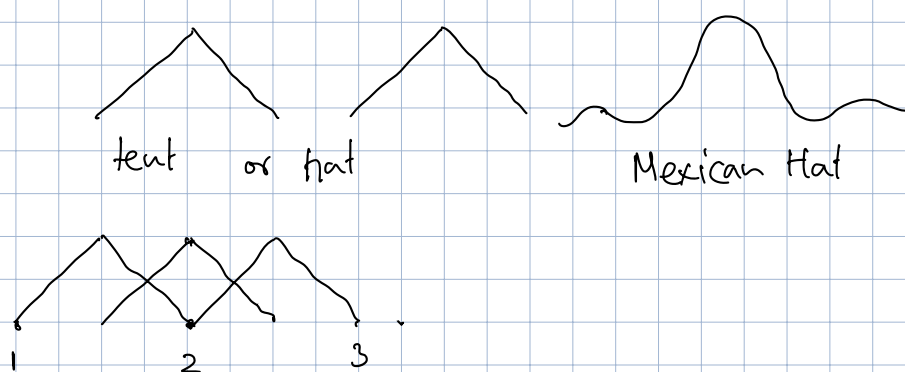
$$(A + M)c = F$$

Remark Any time independent 2nd order PDE will be given by a linear system governed by mass matrix, stiffness matrix, load vector. (M4162/M6201)

Remark: Dimension of the linear system and its stability depends on the choice of basis and test functions
 So we get the best (non-linear) approximation
 (M6210, not in M4162/M6201)

Remark: If the problem is dominantly hyperbolic
 FEM may not be a good choice without "best approximation".

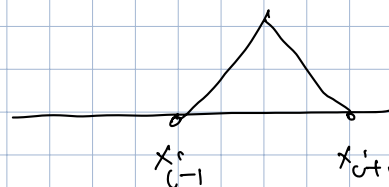
Construction of stiffness matrix using test function.



$$\phi_i(x) = \begin{cases} \frac{x - x_{i-1}^c}{x_i^c - x_{i-1}^c} & x_{i-1}^c \leq x \leq x_i^c \\ \frac{x_{i+1}^c - x}{x_{i+1}^c - x_i^c} & x_i^c \leq x \leq x_{i+1}^c \\ 0 & \text{otherwise} \end{cases}$$

$a(\phi_i, \phi_i)$ diagonal elements of stiffness matrix

$$\begin{aligned} &= \int_0^1 \phi_i'(x) \phi_i'(x) dx = \sum_i \int_{x_{i-1}^c}^{x_{i+1}^c} \phi_i' \phi_i dx \\ &= \int_{x_{i-1}^c}^{x_i^c} \phi_i'(x) \phi_i'(x) dx + \int_{x_i^c}^{x_{i+1}^c} \phi_i'(x) \phi_i'(x) dx \end{aligned}$$



$$= \int_{x_{i-1}^*}^{x_i^*} \frac{1}{x_i^* - x_{i-1}^*} \frac{(t+1)}{x_i^* - x_{i-1}^*} dx + \int_{x_i^*}^{x_{i+1}^*} \frac{-1}{x_{i+1}^* - x_i^*} \frac{-1}{x_{i+1}^* - x_i^*} dx$$

$$= \frac{1}{(x_i^* - x_{i-1}^*)^2} \times \int_{x_{i-1}^*}^{x_i^*} + \frac{1}{(x_{i+1}^* - x_i^*)^2} \times \int_{x_i^*}^{x_{i+1}^*}$$

$$= \frac{x_i^* - x_{i-1}^*}{(x_i^* - x_{i-1}^*)^2} + \frac{x_{i+1}^* - x_i^*}{(x_{i+1}^* - x_i^*)^2} \quad h_i^* = x_i^* - x_{i-1}^*$$

$$= \frac{1}{h_i^*} + \frac{1}{h_{i+1}^*}$$

H.W. $a(\phi_i, \phi_{i+1}) = -\frac{1}{h_{i+1}^*}$

define $\phi_{i+1}(x)$ by shifting index of $\phi_i(x)$

dealing with end point

Recall $u_n = \phi_0 + \sum_{j=1}^n G_j \phi_j$

ϕ_0 satisfies B.C.



ϕ_0 "could" be assigned to node 0

At n -th node

$$\phi_n(x) = \begin{cases} \frac{x - x_{n-1}}{x_n - x_{n-1}} \\ 0 \end{cases}$$

$$x_{n-1} \leq x \leq x_n$$

otherwise

H.W.

$$a(\phi_n, \phi_n) = \frac{1}{h_n}$$

Finally, stiffness matrix for $-u''(x) = f(x)$ using tent functions

$$A = \begin{bmatrix} \frac{1}{h_1} + \frac{1}{h_2} & -\frac{1}{h_2} & & & \\ -\frac{1}{h_2} & \frac{1}{h_2} + \frac{1}{h_3} & -\frac{1}{h_3} & & \\ & & \ddots & \ddots & \\ & & & -\frac{1}{h_n} & \frac{1}{h_n} \end{bmatrix}$$

If $h_1 = h_2 = \dots$

$$A = \frac{1}{h} \begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{bmatrix}$$

Q. If $h_1 = h_2 = \dots = h_n$, for $x \in [0, 1]$
is FEM stiffness matrix same as FDM
(no, but structure is same)

Q. $-u''(x) = f(x)$, $u(0) = 0$, $u(1) = 0$

$h_1 = h_2 = \dots = h_n$

Find discretization matrix using FEM
(not in notes, but notes can be used)

Q. $-\frac{\partial}{\partial x} \left(\nu \frac{\partial u}{\partial x} \right) + bu = f$

Find $\nu a(\phi_i, \phi_j) + b m(\phi_i, \phi_j) = \langle f, \phi_i \rangle$

Q. Derive the weak form of
 $-u''(x) = f(x) \quad 0 < x < 1$
 $u(0) = 0$
 $u'(1) = \beta$

Ans: $\int_0^1 v \frac{\partial u}{\partial x} \frac{\partial v}{\partial x} dx + \int_0^1 b u v dx = \int_0^1 f v dx$

Time dependent PDE (Parabolic)

$$\left. \begin{aligned} \frac{\partial u}{\partial t} &= \frac{\partial^2 u}{\partial x^2} \\ u(a, t) &= 0 \\ u(b, t) &= 0 \\ u(x, 0) &= u_0(x) \end{aligned} \right\} \frac{\partial u}{\partial t} = \frac{\partial}{\partial x} \left(v(x) \frac{\partial u}{\partial x} \right)$$

Idea is

$$\int_a^b \frac{\partial u}{\partial t} v(x) dx = \int_a^b \frac{\partial^2 u}{\partial x^2} v(x) dx$$

$$= \left. \frac{\partial u}{\partial x} v(x) \right|_a^b - \int_a^b \frac{\partial u}{\partial x} \frac{\partial v}{\partial x} dx$$

$$\int u dv = uv - \int v du$$

$$\int u v dx = u \int v dx - \int v \frac{du}{dx} dx$$

$$\int_a^b \frac{\partial u}{\partial t} v dx = - \int_a^b \frac{\partial u}{\partial x} \frac{\partial v}{\partial x} dx + \cancel{\frac{\partial u}{\partial x} v(b) - \frac{\partial u}{\partial x} v(a)}$$

because test function should satisfy B.C.

$$\frac{\partial}{\partial t} \int_a^b u v dx = - \int_a^b \frac{\partial u}{\partial x} \frac{\partial v}{\partial x} dx$$

We have two bilinear forms

$$m(u, v) = \int_a^b uv \, dx$$

$$a(u, v) = \int_a^b \frac{\partial u}{\partial x} \frac{\partial v}{\partial x} \, dx$$

Subspace

$$u(x, t) \approx u_h = \sum_{j=1}^n c_j(t) \phi_j(x), \quad v(x) = \phi_i, \quad i=1, \dots, n$$

$$\frac{\partial}{\partial t} \int_a^b \left(\sum_{j=1}^n c_j \phi_j \right) \phi_i \, dx = - \int_a^b \left(\sum_{j=1}^n c_j \phi_j' \right) \phi_i \, dx$$

$$= - \sum_{j=1}^n c_j(t) \int_a^b \phi_j' \phi_i \, dx$$

$$\sum_{j=1}^n \frac{\partial c_j}{\partial t} \int_a^b \phi_j \phi_i \, dx = - \sum_{j=1}^n c_j(t) \int_a^b \phi_j' \phi_i \, dx$$

Can be written as

$$M \frac{dc}{dt} = -Ac$$

$$M = [m(\phi_i, \phi_j)], \quad A = [a(\phi_i, \phi_j)], \quad c = [c_j]$$
